PRACTICE TASK 1
Data preparation and cleaning

Step 1: getting the raw data

Date	Client	Contact	Department	Payment	Revenue	Profit	Profit Margin
######	AMAZON.CO	Bill SmITH	Cloud Tech_Texas	Transfer	$4,500	$598	13.3%
######	TESLA, INC. (:	KEN Singh	Strategy_New York	PayPal	$3,800	$1,045	27.5%
######	NETFLIX, INC.	Harley I	Strategy_New York		$3,713	$1,009	27.2%
######	THE PROCTEI	Nyla Novak	Operations_Florida			$779	#DIV/0!
######	THE GOLDMA	David Rasn	Operations_Florida	Check	$5,000	$684	13.7%
######	JPMORGAN C	IVAN HINEY	Cloud Tech_Texas	Transfer	$6,100	$544	8.9%
######	MORGAN STA	JONha Ma	Cloud Tech_Texas	Transfer	$4,625	$670	14.5%
######	CITIGROUP IN	Jordan Bo	Cloud Tech_Texas	Transfer	$3,800	$2,045	53.8%
######	BANK OF AM	Kylee Townse	Cloud Tech_Texas	Card	$3,600	$1,564	43.4%
######	WALMART IN	Nora Rollins	Cloud Tech_Texas	Check	$5,100	$1,220	23.9%
######	TARGET COR	BRENDAN W	Cloud Tech_Texas	Check	$4,750	$1,435	30.2%
######	COSTCO WH	Conor Wis	Operations_Florida	Transfer	$6,000	$998	16.6%
######	MCDONALD'S	Steven MIC	Big Data_California	Check	$4,500	$780	17.3%
######	EXXON MOBI	Lucia Mckay	Big Data_California	Card		$1,044	#DIV/0!
######	VERIZON CO	JoSE Roach	Big Data_California	Transfer	$3,713	$1,222	32.9%
######	THE HOME D	Franklin WRI	Big Data_California	Transfer	$4,950	$1,065	21.5%
######	CISCO SYSTEI	Alia Th	Operations_Florida	Transfer	$4,750	$810	17.1%
######	CHEVRON CO	Denzel F	Operations_Florida	Transfer	$7,320	$933	12.7%
######	COSTCO WH	Conor Wis	Operations_Florida	Transfer	$6,000	$998	16.6%
######	MCDONALD'S	Steven MIC	Big Data_California	Check	$4,500	$780	17.3%
######	AT&T INC. (X	Bruno Cordo	Big Data_California	Transfer	$5,088	$655	12.9%
######	INTEL CORPC	jaylynn nap	Big Data_California	Transfer	$4,500	$722	16.0%
######	GENERAL MC	Bruce RICH	Big Data_California	Card	$4,250	$901	21.2%
######	MICROSOFT	Arturo Mo	Big Data_California	PayPal	$5,250	$1,349	25.7%
######	COMCAST CO	Bryce Carp	Strategy_New York	PayPal	$6,500	$1,288	19.8%
######	DELL TECHN	Jaidyn ANDEF	Strategy_New York	PayPal	$7,500	$1,664	22.2%
######	JOHNSON & .	MARK WALM	Strategy_New York	Transfer	$5,500	$1,320	24.0%
######	FEDEX CORP	HARRY LEE	Strategy_New York	Transfer	$4,625	$1,001	21.6%
######	GENERAL ELE	JOSH JOHNSC	Strategy_New York	Transfer	$4,500	$960	21.3%
######	LOCKHEED M	Mik Naam	Strategy_New York	Card	$5,400	$540	10.0%
######	THE GOLDMA	David Rasn	Operations_Florida	Check	$5,000	$684	13.7%

Step 2: cleaning the raw data

CS301 REVENUE AND PROFIT INFORMATION								
Date	**Client**	**Contacts**	**Department**	**State**	**Payment**	**Revenue**	**Profit**	**Profit Margin**
30/05/2023	AMAZON.COM, INC.	Bill Smith	Cloud Tech	Texas	Transfer	$ 4,500.00	$ 598.00	13%
30/05/2023	TESLA, INC.	Ken Singh	Strategy	New York	PayPal	$ 3,800.00	$1,045.00	28%
30/05/2023	NETFLIX, INC.	Harley Fritz	Strategy	New York	Others	$ 3,712.50	$1,009.00	27%
30/05/2023	THE PROCTER & GAMBLE COMPANY	Nyla Novak	Operations	Florida	Others	$ 4,994.00	$ 779.00	16%
30/05/2023	THE GOLDMAN SACHS GROUP, INC.	David Rasmussen	Operations	Florida	Check	$ 5,000.00	$ 684.00	14%
31/05/2023	JPMORGAN CHASE & CO.	Ivan Hiney	Cloud Tech	Texas	Transfer	$ 6,100.00	$ 544.00	9%
31/05/2023	MORGAN STANLEY	Jonha Ma	Cloud Tech	Texas	Transfer	$ 4,625.00	$ 670.00	14%
31/05/2023	CITIGROUP INC.	Jordan Boone	Cloud Tech	Texas	Transfer	$ 3,800.00	$2,045.00	54%
31/05/2023	BANK OF AMERICA CORPORATION	Kylee Townsend	Cloud Tech	Texas	Card	$ 3,600.00	$1,564.00	43%
31/05/2023	WALMART INC.	Nora Rollins	Cloud Tech	Texas	Check	$ 5,100.00	$1,220.00	24%
31/05/2023	TARGET CORPORATION	Brendan Wallace	Cloud Tech	Texas	Check	$ 4,750.00	$1,435.00	30%
31/05/2023	COSTCO WHOLESALE CORPORATION	Conor Wise	Operations	Florida	Transfer	$ 6,000.00	$ 998.00	17%
31/05/2023	MCDONALD'S CORPORATION	Steven Michael	Big Data	California	Check	$ 4,500.00	$ 780.00	17%
01/06/2023	EXXON MOBIL CORPORATION	Lucia Mckay	Big Data	California	Card	$ 4,994.00	$1,044.00	21%
01/06/2023	VERIZON COMMUNICATIONS INC.	Jose Roach	Big Data	California	Transfer	$ 3,712.50	$1,222.00	33%
01/06/2023	THE HOME DEPOT, INC.	Franklin Wrigt	Big Data	California	Transfer	$ 4,950.00	$1,065.00	22%
01/06/2023	CISCO SYSTEMS, INC.	Alia Thornton	Operations	Florida	Transfer	$ 4,750.00	$ 810.00	17%
01/06/2023	CHEVRON CORPORATION	Denzel Flores	Operations	Florida	Transfer	$ 7,320.00	$ 933.00	13%
01/06/2023	AT&T INC.	Bruno Cordova	Big Data	California	Transfer	$ 5,087.50	$ 655.00	13%
01/06/2023	INTEL CORPORATION	Jaylynn Napp	Big Data	California	Transfer	$ 4,500.00	$ 722.00	16%
01/06/2023	GENERAL MOTORS COMPANY	Bruce Rich	Big Data	California	Card	$ 4,250.00	$ 901.00	21%
02/06/2023	MICROSOFT CORPORATION	Arturo Moore	Big Data	California	PayPal	$ 5,250.00	$1,349.00	26%
02/06/2023	COMCAST CORPORATION	Bryce Carpenter	Strategy	New York	PayPal	$ 6,500.00	$1,288.00	20%
02/06/2023	DELL TECHNOLOGIES INC.	Jaidyn Andersen	Strategy	New York	PayPal	$ 7,500.00	$1,664.00	22%
02/06/2023	JOHNSON & JOHNSON	Mark Walm	Strategy	New York	Transfer	$ 5,500.00	$1,320.00	24%
02/06/2023	FEDEX CORPORATION	Harry Lee	Strategy	New York	Transfer	$ 4,625.00	$1,001.00	22%
02/06/2023	GENERAL ELECTRIC COMPANY	Josh Johnson	Strategy	New York	Transfer	$ 4,500.00	$ 960.00	21%
02/06/2023	LOCKHEED MARTIN CORPORATION	Mik Naam	Strategy	New York	Card	$ 5,400.00	$ 540.00	10%

Step 3: determining the fact contents of the fact table and its dimensions

For this step as it was a Practice task we were following what content should be in the fact table and what content should be in each of its dimension tables.

Step 4: creating the dimension tables

Client dimension table

Client	client_id	Contacts
AMAZON.COM, INC.	c-101	Bill Smith
AT&T INC.	c-102	Bruno Cordova
BANK OF AMERICA CORPORATION	c-103	Kylee Townsend
CHEVRON CORPORATION	c-104	Denzel Flores
CISCO SYSTEMS, INC.	c-105	Alia Thornton
CITIGROUP INC.	c-106	Jordan Boone
COMCAST CORPORATION	c-107	Bryce Carpenter
COSTCO WHOLESALE CORPORATION	c-108	Conor Wise
DELL TECHNOLOGIES INC.	c-109	Jaidyn Andersen
EXXON MOBIL CORPORATION	c-110	Lucia Mckay
FEDEX CORPORATION	c-111	Harry Lee
GENERAL ELECTRIC COMPANY	c-112	Josh Johnson
GENERAL MOTORS COMPANY	c-113	Bruce Rich
INTEL CORPORATION	c-114	Jaylynn Napp
JOHNSON & JOHNSON	c-115	Mark Walm
JPMORGAN CHASE & CO.	c-116	Ivan Hiney
LOCKHEED MARTIN CORPORATION	c-117	Mik Naam
MCDONALD'S CORPORATION	c-118	Steven Michael
MICROSOFT CORPORATION	c-119	Arturo Moore
MORGAN STANLEY	c-120	Jonha Ma
NETFLIX, INC.	c-121	Harley Fritz
TARGET CORPORATION	c-122	Brendan Wallace
TESLA, INC.	c-123	Ken Singh
THE GOLDMAN SACHS GROUP, INC.	c-124	David Rasmussen
THE HOME DEPOT, INC.	c-125	Franklin Wrigt
THE PROCTER & GAMBLE COMPANY	c-126	Nyla Novak
VERIZON COMMUNICATIONS INC.	c-127	Jose Roach
WALMART INC.	c-128	Nora Rollins

Department dimension table

Department	dept_id
Big Data	d-101
Cloud Tech	d-102
Operations	d-103
Strategy	d-104

State dimension table

State	state_id
California	s-101
Florida	s-102
New York	s-103
Texas	s-104

Payment dimension table

Payment	payment_id
Card	p-101
Check	p-102
Others	p-103
PayPal	p-104
Transfer	p-105

Step 5: creating the fact table

Trans_id	Date	Client_id	Dept_id	State_id	Payment_id	Revenue	Profit	Profit Margin
T-101	30/05/2023	c-101	d-102	s-104	p-105	$ 4,500.00	$ 598.00	13%
T-102	30/05/2023	c-123	d-104	s-103	p-104	$ 3,800.00	$ 1,045.00	28%
T-103	30/05/2023	c-121	d-104	s-103	p-103	$ 3,712.50	$ 1,009.00	27%
T-104	30/05/2023	c-126	d-103	s-102	p-103	$ 4,994.00	$ 779.00	16%
T-105	30/05/2023	c-124	d-103	s-102	p-102	$ 5,000.00	$ 684.00	14%
T-106	31/05/2023	c-116	d-102	s-104	p-105	$ 6,100.00	$ 544.00	9%
T-107	31/05/2023	c-120	d-102	s-104	p-105	$ 4,625.00	$ 670.00	14%
T-108	31/05/2023	c-106	d-102	s-104	p-105	$ 3,800.00	$ 2,045.00	54%
T-109	31/05/2023	c-103	d-102	s-104	p-101	$ 3,600.00	$ 1,564.00	43%
T-110	31/05/2023	c-128	d-102	s-104	p-102	$ 5,100.00	$ 1,220.00	24%
T-111	31/05/2023	c-122	d-102	s-104	p-102	$ 4,750.00	$ 1,435.00	30%
T-112	31/05/2023	c-108	d-103	s-102	p-105	$ 6,000.00	$ 998.00	17%
T-113	31/05/2023	c-118	d-101	s-101	p-102	$ 4,500.00	$ 780.00	17%
T-114	01/06/2023	c-110	d-101	s-101	p-101	$ 4,994.00	$ 1,044.00	21%
T-115	01/06/2023	c-127	d-101	s-101	p-105	$ 3,712.50	$ 1,222.00	33%
T-116	01/06/2023	c-125	d-101	s-101	p-105	$ 4,950.00	$ 1,065.00	22%
T-117	01/06/2023	c-105	d-103	s-102	p-105	$ 4,750.00	$ 810.00	17%
T-118	01/06/2023	c-104	d-103	s-102	p-105	$ 7,320.00	$ 933.00	13%
T-119	01/06/2023	c-102	d-101	s-101	p-105	$ 5,087.50	$ 655.00	13%
T-120	01/06/2023	c-114	d-101	s-101	p-105	$ 4,500.00	$ 722.00	16%
T-121	01/06/2023	c-113	d-101	s-101	p-101	$ 4,250.00	$ 901.00	21%
T-122	02/06/2023	c-119	d-101	s-101	p-104	$ 5,250.00	$ 1,349.00	26%
T-123	02/06/2023	c-107	d-104	s-103	p-104	$ 6,500.00	$ 1,288.00	20%
T-124	02/06/2023	c-109	d-104	s-103	p-104	$ 7,500.00	$ 1,664.00	22%
T-125	02/06/2023	c-115	d-104	s-103	p-105	$ 5,500.00	$ 1,320.00	24%
T-126	02/06/2023	c-111	d-104	s-103	p-105	$ 4,625.00	$ 1,001.00	22%
T-127	02/06/2023	c-112	d-104	s-103	p-105	$ 4,500.00	$ 960.00	21%
T-128	02/06/2023	c-117	d-104	s-103	p-101	$ 5,400.00	$ 540.00	10%